

PRAJWAL KUMAR K

Bengaluru, India | +91-9740167850
prajwal.k141205@gmail.com | [LinkedIn](#) | [Portfolio](#)
[GitHub](#) | [Codeforces](#) | [CodeChef](#) | [LeetCode](#)

EDUCATION

Indian Institute of Information Technology, Kottayam
B.Tech in Computer Science and Engineering; CGPA: 8.66/10.0

Kottayam, Kerala
Aug 2024 – May 2028 (expected)

RESEARCH INTERESTS

Algorithms and systems, game theory, machine learning and data science, AI safety, and NLP.

SKILLS

Programming: Python, C++, Rust, MATLAB
ML and data: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, Matplotlib
Research tools: Jupyter, Simulink, FastAPI, Next.js, SQL
Current study: ISLP, Mathematics for Machine Learning, Stanford CS231n (ongoing)
Spoken languages: English, Kannada, German and Hindi (fluent)

EXPERIENCE AND VOLUNTEERING

Technical Lead, Openverse (Open Source Club) May 2025 – Present
IIIT Kottayam Kottayam, Kerala
– Lead technical work and PR reviews; contribute sorting visualizations to the team-built Algorithmia platform.
Coordinator, Coders Club Jan 2025 – Present
IIIT Kottayam Kottayam, Kerala
– Curate problems, configure Codeforces Gym rounds and help run campus programming contests.

RESEARCH AND SELECTED PROJECTS

PokeForge | *Gen 9 OU Battle Agents, Python, PyTorch* GitHub | In progress
– Reproduced pinned FoulPlay and Metamon baselines in isolated simulator environments; testing simulator conformance and agent strength before drawing performance conclusions.
Bluff | *Repeated-Play Opponent Modeling* GitHub
– Built an imperfect-information card-game simulator where agents lie and call bluffs; persistent Bayesian profiles model recurring opponents across rounds.
– Across eight synthetic profiles, removing the learned bluff detector reduced draw-aware score by 4.9–5.8 percentage points; this result is limited to the simulator and tested roster.
Drishti-XAI | *Retinal Imaging, MATLAB, Python* SIH 2026 submission
– Built the MATLAB/Python prototype and evaluation pipeline for explainable retinal screening, covering image-quality checks, grading, segmentation, Grad-CAM, calibration and reporting.
– At a fixed model/threshold, internal APTOS (n=733) achieved 0.908 sensitivity/0.877 specificity; external Messidor-2 achieved 0.363/0.865 with AUC 0.666, exposing a transfer gap.
KPlaceNet | *Image Geolocation, PyTorch, ResNet-50* GitHub
– Evaluated quadtree-based CNN geolocation; the best of 12 ResNet-50 configurations placed 3.33% of 3,000 spatially separated OSV-5M test images within 200 km.

PUBLICATION AND ACHIEVEMENTS

K. Prajwal Kumar, Rubiya Subair, and Shajulin Benedict. “Collaboration-Enabled Energy-Aware Learning Model Exchange (CE-LME) Framework for Edge Computing-Assisted Applications.” *2025 IEEE 3rd International Conference on Inventive Computing and Informatics*, pp. 1390–1395. DOI. First-listed author; conference presenter. Algorithmia selected for OOSC 4.0; team lightning talk at OOSC 3.0 and presentation at T4E 2025, IIT Madras. Best Software Award, IEEE MACE .hack(); '26, Team Hogorithm.

OPEN-SOURCE CONTRIBUTION

PettingZoo Leduc Hold'em (PR #1500, open): migrated the environment to an OpenSpiel-backed Shimmy adapter; focused tests passed.